A Diagrammatic Proof of the Haldane Gap via Spacetime Tensor Networks

(Dated: October 8, 2026)

Following the recent machine-assisted resolution of the Haldane conjecture for periodic and boundary-field spin-1 Heisenberg chains by OpenAI [1, 2], we show that the analytical derivations can be recast as elementary tensor-network contractions. Formulating the system as an exact continuous-time thermal matrix product operator exposes the underlying spacetime geometry. The step from a finite-scale estimate to a uniform gap then reduces to a contractive scale-doubling recurrence on planar diagrams, under three pages long and formalized in Lean 4 with the two numerical inputs as its only unproved hypotheses. The finite-scale estimate itself is taken unchanged from Ref. [1], whose certificate uses spectral filters and character projections. This illustrates how diagrammatic representations can clarify and compress machine-generated proofs in quantum many-body physics.

We consider the antiferromagnetic spin-1 Heisenberg chain with periodic boundary conditions $\boldsymbol{S}_{L+1} = \boldsymbol{S}_1$ and $J = k_B = 1$:

$$H_L = \sum_{j=1}^{L} \boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1}, \qquad Z_L(\beta) = \operatorname{Tr} e^{-\beta H_L}. \tag{1}$$

All system sizes are even and at least four. Let $E_0(L) \le E_1(L) \le \cdots$ include multiplicities and define the gap $\Delta_L = E_1(L) - E_0(L)$, the step from the ground level to the first excited level,  . For the Gibbs weights $w_m = e^{-\beta E_m(L)}/Z_L(\beta)$,

$$\frac{w_1}{w_0} = e^{-\beta\Delta_L} \le e^{-c\beta} \quad \Longrightarrow \quad \Delta_L \ge c. \tag{2}$$

The ratio decays in β like  ; the task is to make its rate c independent of L:

Theorem 1 (Periodic gap [1])**.** *For every even $L \ge 2304$, the chain (1) has a unique ground state and $\Delta_L > (\log 20)/784 \simeq 0.003821$.*

The spectral gap of the chain (1) was established by OpenAI [1] using machine-assisted proof search over a thirty-page analytical construction. That proof relies on coordinate expansions, degree-12 polynomial spectral filters, and character decompositions across dihedral symmetry sectors. Automated provers tend to work in such coordinate representations, and over long chains of deductions these accumulate intermediate algebra.

Here, we show that the underlying mechanism is fundamentally geometric. The proof has one goal and three steps. The goal is a single number per length and temperature: the *thermal purity* of the Gibbs state, which is close to one exactly when the ground state carries almost all of the Boltzmann weight. A thermal purity near one forces a gap, so it suffices to show that the purity defect falls exponentially in β, uniformly in L. Squaring the Gibbs state in imaginary time improves the purity for any Hamiltonian, but it never changes the length. The three steps supply the missing direction. First, an exact continuous-time thermal MPO turns the partition function into a power of a symmetric transfer operator in space, so a *spatial purity* can be squared in the same way. Second, spacetime contraction identities couple the spatial and thermal purities, and doubling the length and the inverse temperature together roughly squares both defects; we call this self-improving iteration the *spacetime bootstrap*. Third, monotonicity of ℓ^p norms extends the bound from the doubled lengths to all intermediate even lengths, which establishes Eq. (2).

Every step below is a contraction of one tensor network: horizontal legs — are virtual (space), vertical double legs ‖ are physical (imaginary time), joined legs are contracted, closed loops  are traces [3, 4], and a labeled box  is a product of tensors. Blue marks thermal objects ρ and red marks spatial ones σ. Appendix A applies the same networks to an open chain with boundary fields, where the spacetime is a cylinder rather than a torus; it reuses every identity below and adds only boundary vectors.

From purity to gap.— We first show why a purity close to one forces a gap. For a positive trace-one operator with eigenvalues $p_0 \ge p_1 \ge \cdots$, the purity obeys

$$P = \sum_i p_i^2 \le p_0 \sum_i p_i = p_0. \tag{3}$$

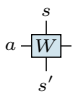
Thus $1 - p_0 \le 1 - P$: a purity near one forces a spectrum like  with a single dominant bar. For the Gibbs state, whose eigenvalues are the weights w_m of Eq. (2), $P > 1/2$ implies a unique ground state, with excited-state population $1 - w_0 \le 1 - P$, and hence $e^{-\beta\Delta_L} = w_1/w_0 \le (1-P)/P$. To reach Eq. (2) it therefore suffices to show that the defect $1 - P$ falls like $e^{-c\beta}$, with the same c for every L. The difficulty is the length: at fixed β, a longer ring has more thermal excitations and a smaller purity, so L and β must grow together. Imaginary time already has a squaring operation, Eq. (11) below; space needs one too, and this requires a transfer operator.

Thermal MPO.— For the classical Ising chain, $Z_n = \operatorname{Tr} K^n$ with a 2×2 transfer matrix K that steps one site along the ring. We construct the quantum counterpart: a real symmetric operator X_β with $Z_n(\beta) = \operatorname{Tr}_{\mathrm{v}} X_\beta^n$. To obtain a real symmetric transfer operator, use the Cartesian spin basis $s, s' \in \{x, y, z\}$. The matrices $(S^\alpha)_{ss'} = -i\epsilon_{\alpha ss'}$ give real operators $G_\alpha = iS^\alpha$, $\|G_\alpha\| = 1$, and $-H_L = \sum_{j,\alpha} G_{\alpha,j} G_{\alpha,j+1}$. The variable τ below is imaginary time. Because the ordered region $0 < \tau_1 < \cdots < \tau_k < \beta$ has volume $\beta^k/k!$, the exponen-

tial series of any matrix A can be written as a sum over ordered event times,

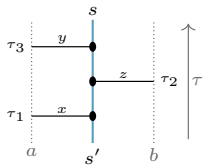
$$e^{\beta A} = \sum_{k\geq 0} \int_{0<\tau_1<\cdots<\tau_k<\beta} d\tau_1 \cdots d\tau_k \; \underbrace{A \cdots A}_{k}. \tag{4}$$

For $A = -H_L$, each factor selects one bond and one label α, so each term of Eq. (4) is a set of *bond events* (τ, α), drawn as rungs between the worldlines of neighboring sites. Each bond event (τ, α) in the exponential expansion inserts G_α at both endpoints. Let a, b record these events on the two bonds adjacent to a site, with $0 < \tau < \beta$. The local MPO tensor is



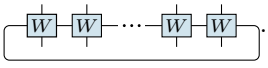
$$= W_\beta^{ss'}(a,b) = \left\langle s \middle| \mathcal{T}_\tau \prod_{(\tau,\alpha)\in a\cup b} G_\alpha \middle| s' \right\rangle, \tag{5}$$

where $\mathcal{T}_\tau$ places later times to the left. Inside the box, W is the site's worldline read from bottom to top: for $a = \{(\tau_1, x), (\tau_3, y)\}$, $b = \{(\tau_2, z)\}$ and $\tau_1 < \tau_2 < \tau_3$, the tensor $W_\beta^{ss'}(a,b)$ is



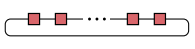
$$= \langle s | G_y G_z G_x | s' \rangle. \tag{6}$$

Operators on different sites commute, so once the events on every bond are fixed, the time-ordered product over the ring splits into one such worldline per site. The events on a bond therefore play the role of the Ising spin between two copies of K, with the 2×2 matrix K replaced by an operator on event histories. Summing event labels and integrating ordered times defines a measure μ_β of mass $e^{3\beta}$, with unit mass on the empty history. Contracting n tensors around the ring recovers the absolutely convergent exponential series as an exact MPO on the virtual space $L^2(\mu_\beta)$, without bond truncation [3, 4]:



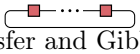
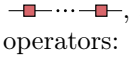
$$e^{-\beta H_n} = \tag{7}$$

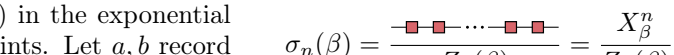
Tracing the physical index,  = , gives the spatial transfer kernel $X_\beta(a,b) = \sum_s W_\beta^{ss}(a,b)$. Since a and b enter Eq. (5) only through their union, $X_\beta(a,b) = X_\beta(b,a)$. The classical K has this left-right symmetry for free; for the quantum chain it holds because each event inserts the same real matrix at both ends of its bond. This is the *reflection positivity* of the bond, $-\boldsymbol{S}_j \cdot \boldsymbol{S}_{j+1} = \sum_\alpha G_{\alpha,j} G_{\alpha,j+1}$; for other Hamiltonians it is a genuine condition [5]. The kernel is real, symmetric, and bounded in absolute value by three (the trace of a product of 3×3 matrices of norm one); since μ_β has finite mass, X_β is self-adjoint and Hilbert–Schmidt on $L^2(\mu_\beta)$. Closing the ring of transfer vertices gives, for even n,



$$Z_n(\beta) = = \mathrm{Tr}_{\mathrm{v}}\, X_\beta^n = \sum_i |\lambda_i(\beta)|^n, \tag{8}$$

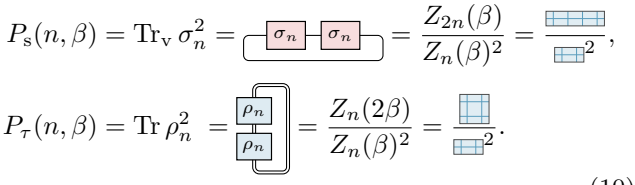
where $\lambda_i(\beta)$ are the real eigenvalues of X_β, ordered by modulus, $|\lambda_0| \geq |\lambda_1| \geq \cdots$. Raising them to the nth power concentrates the weight on the largest: for $n = 4$.

Spatial and thermal purities.— The transfer operator gives space its own density operator. Cut the ring of Eq. (8) at one bond,  $\mapsto$  , and normalize both the transfer and Gibbs operators:



$$\sigma_n(\beta) = = \frac{X_\beta^n}{Z_n(\beta)}, \qquad \rho_n(\beta) = \frac{e^{-\beta H_n}}{Z_n(\beta)}. \tag{9}$$

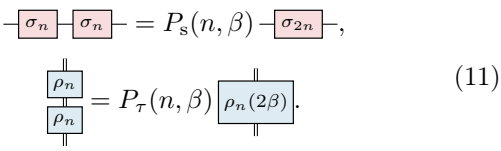
For even n, both operators are positive with unit trace; denote their eigenvalues by p_i and w_m, respectively. Their purities join two copies, along space for σ_n and along imaginary time for ρ_n. Drawing $Z_n(\beta)$ as a space-time torus  of width n and height β (dashed edges identified), both are one ratio of tori:



$$P_{\mathrm{s}}(n,\beta) = \mathrm{Tr}_{\mathrm{v}}\, \sigma_n^2 = = \frac{Z_{2n}(\beta)}{Z_n(\beta)^2} = ,$$

$$P_\tau(n,\beta) = \mathrm{Tr}\, \rho_n^2 = = \frac{Z_n(2\beta)}{Z_n(\beta)^2} = . \tag{10}$$

Squaring.— Squaring is the operation that improves a purity. Squaring σ_n doubles the length; squaring ρ_n doubles the inverse temperature:



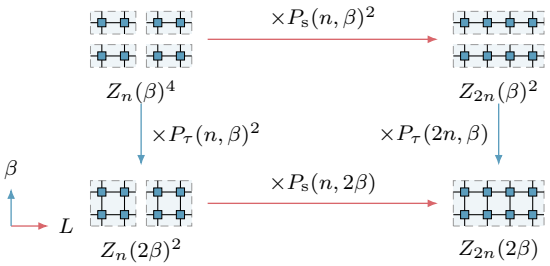
(11)

Both replace the normalized eigenvalues p_i by p_i^2/P, which concentrates the spectrum:  $\mapsto$  . To bound the new purity P', use $1 - P = 2\sum_{i<k} p_i p_k$:

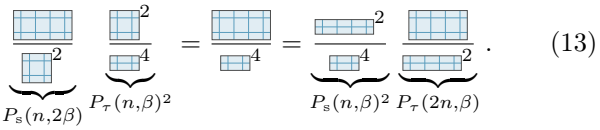
$$1 - P' = \frac{2\sum_{i<k} p_i^2 p_k^2}{P^2} \leq \frac{(1-P)^2}{2P^2} = f(1-P),$$
$$f(z) = \frac{z^2}{2(1-z)^2}. \tag{12}$$

Dropping cross terms from $(\sum_{i<k} p_i p_k)^2$ gives the inequality. The function f increases on $[0,1)$, with $f(z) \sim z^2/2$: squaring roughly squares the purity defect $1 - P$, keeping it far below the diagonal,  for $0 \leq z \leq .3$.

Spacetime coupling identities.— Each squaring doubles one side of the torus: space ($n \to 2n$),  $\mapsto$  , or imaginary time ($\beta \to 2\beta$),  $\mapsto$  . As explained above, a uniform gap needs both sides to grow together, so we need identities that trade a purity in one direction for a purity in the other. Contracting four copies of the partition function $Z_n(\beta)$ in either spatial or temporal order yields the composite torus $Z_{2n}(2\beta)$. The networks are drawn to scale; red arrows join copies in space, blue arrows in time, and each arrow carries the purity factor it multiplies by:



Writing each purity as its ratio of tori, Eq. (10), the two paths cancel to the same quotient:



(13)

Solving for one factor, and repeating at length $2n$, gives

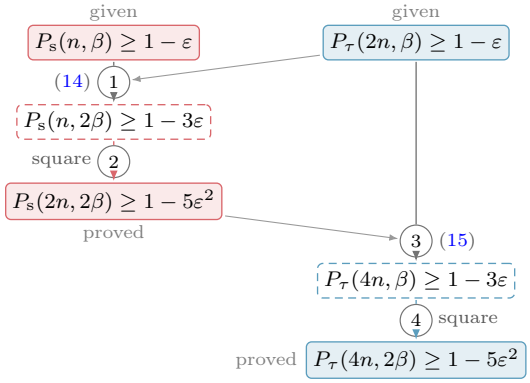
$$P_s(n,2\beta) = \frac{P_s(n,\beta)^2 P_\tau(2n,\beta)}{P_\tau(n,\beta)^2}, \tag{14}$$

$$P_\tau(4n,\beta) = \frac{P_\tau(2n,\beta)^2 P_s(2n,2\beta)}{P_s(2n,\beta)^2}. \tag{15}$$

Doubling estimate.— Alternating the two coupling identities with the two squarings doubles the torus in both directions,  , and roughly squares the purity defect. For even n and $0 \le \varepsilon \le 1/100$,

$$\begin{aligned} &P_s(n,\beta),\ P_\tau(2n,\beta) \ge 1-\varepsilon \\ \Longrightarrow\quad &P_s(2n,2\beta),\ P_\tau(4n,2\beta) \ge 1-5\varepsilon^2. \end{aligned} \tag{16}$$

The scale-doubling update (16) is established by four moves, alternating between spacetime coupling and operator squaring. Read the diagram below from top to bottom: each box is one bound, red for P_s and blue for P_τ; solid boxes are given or proved, dashed boxes intermediate. Each move is a numbered circle labeled by the identity it uses; the grey arrow entering a circle is the second bound that move needs:



The proved pair is the given pair with $(n,\beta) \to (2n,2\beta)$, so it is the given pair of the next scale, which is why the estimate can be iterated.

Move 1 (coupling). Since $P_\tau(n,\beta) \le 1$, Eq. (14) gives $P_s(n,2\beta) \ge (1-\varepsilon)^3 \ge 1-3\varepsilon$. *Move 2* (squaring in space). For even n, $\sigma_n(2\beta)$ has nonnegative spectrum and squares to $\sigma_{2n}(2\beta)$, so Eq. (12) and monotonicity of f give $P_s(2n,2\beta) \ge 1-\delta$, where

$$\delta = f\big(1-(1-\varepsilon)^3\big) = \tfrac{1}{2}\big[(1-\varepsilon)^{-3}-1\big]^2 \le 5\varepsilon^2 \le \varepsilon. \tag{17}$$

Here $1-(1-\varepsilon)^3 \le 3\varepsilon$ and $9/[2(1-3\varepsilon)^2] \le 5$ for $\varepsilon \le 1/100$. *Move 3* (coupling). Eq. (15) with $P_s(2n,\beta) \le 1$ and $\delta \le \varepsilon$ gives $P_\tau(4n,\beta) \ge (1-\varepsilon)^2(1-\delta) \ge (1-\varepsilon)^3$. *Move 4* (squaring in time). $\rho_{4n}(\beta)$ squares to $\rho_{4n}(2\beta)$, so $P_\tau(4n,2\beta) \ge 1-\delta \ge 1-5\varepsilon^2$. This proves Eq. (16).

Starting estimate and iteration.— The doubling estimate needs a starting point: one scale at which both defects are already below $1/100$. This is the only numerical input. Proposition 5.5 of Ref. [1] provides the starting estimates:

$$P_s(2304,784),\ P_\tau(4608,784) \ge 0.99. \tag{18}$$

Ref. [1] shifts H_n by a constant per site; this multiplies $Z_n(\beta)$ by $e^{-\kappa\beta n}$ for some κ, which cancels from both purity ratios of Eq. (10). After $j \ge 0$ doublings, set

$$n_j = 2304\,2^j, \qquad \beta_j = 784\,2^j, \qquad \varepsilon_j = \frac{20^{-2^j}}{5}. \tag{19}$$

Since $\varepsilon_0 = 1/100$ and $\varepsilon_{j+1} = 5\varepsilon_j^2$, Eq. (16) gives $P_s(n_j,\beta_j), P_\tau(2n_j,\beta_j) \ge 1-\varepsilon_j$ by induction on j: each application runs the four moves above once. The defect falls doubly exponentially in j, $\varepsilon_0 = 10^{-2} \to \varepsilon_4 \approx 3\times10^{-22}$ (log scale), and so exponentially in inverse temperature, $\varepsilon_j = \frac{1}{5}\exp[-\beta_j(\log 20)/784]$, as needed in Eq. (2).

All even lengths.— These estimates hold only at the doubled lengths n_j and $2n_j$. To cover every even $L \ge 2304$, choose j with $n_j \le L \le 2n_j$ and compare L with its two neighbors. By Eq. (8), $Z_L(\beta)^{1/L}$ is the ℓ^L norm of the eigenvalue sequence $(|\lambda_i(\beta)|)_i$, and ℓ^p norms decrease as p grows. Applied with $n_j \le L$ at β_j and with $L \le 2n_j$ at $2\beta_j$, this monotonicity gives

$$\begin{aligned} Z_L(\beta_j) &\le Z_{n_j}(\beta_j)^{L/n_j}, \\ Z_L(2\beta_j) &\ge Z_{2n_j}(2\beta_j)^{L/(2n_j)}. \end{aligned} \tag{20}$$

Substitution into the thermal-purity ratio, together with $Z_{2n}(2\beta)/Z_n(\beta)^4 = P_\tau(2n,\beta)P_s(n,\beta)^2$ from Eq. (10), gives

$$\begin{aligned} P_\tau(L,\beta_j) &\ge \big[P_\tau(2n_j,\beta_j)P_s(n_j,\beta_j)^2\big]^{L/(2n_j)} \\ &\ge (1-\varepsilon_j)^3 \ge 1-3\varepsilon_j > \tfrac{1}{2}. \end{aligned} \tag{21}$$

Here $L/(2n_j) \le 1$, and both purities in brackets are at least $1-\varepsilon_j$.

Recovering the gap.— We now return to the argument that opened the proof, with the purity bound (21) as input. Order the eigenvalues of $\rho_L(\beta_j)$ as $w_0 \ge w_1 \ge \cdots$. By Eqs. (3) and (21), $w_0 \ge 1-3\varepsilon_j > 1/2$: the Gibbs spectrum looks like  , with the ground state alone above one half. Two ground states would show two equal top bars,  , neither above one half; so the ground state is unique. The remaining weight bounds w_1, and hence

$$e^{-\beta_j\Delta_L} = \frac{w_1}{w_0} \le \frac{1-w_0}{w_0} \le \frac{3\varepsilon_j}{1-3\varepsilon_j} < 20^{-2^j}. \tag{22}$$

The last inequality uses $3\varepsilon_j/(1-3\varepsilon_j) \le (60/97)20^{-2^j}$. Taking logarithms and dividing by $\beta_j = 784\,2^j$ gives, for every even $L \ge 2304$,

$$\boxed{\Delta_L > \frac{\log 20}{784} \simeq 0.003821.} \tag{23}$$

This proves Theorem 1, conditional on Eq. (18); in particular $\liminf_{L\to\infty,\, L\,\text{even}} \Delta_L \ge (\log 20)/784 > 0$. The scale-doubling recurrence and the gap deduction are formally verified in Lean 4 (`lean/CertifiedGaps/PurityCriterion.lean`).

DISCUSSION: REPRESENTATION IN MACHINE-ASSISTED PROOFS

The uniform spectral gap theorems for the periodic and boundary-field spin-1 Heisenberg chains, together with the certified starting estimates, come from the machine-assisted proofs of OpenAI [1, 2].

Those proofs run to thirty pages. They locate eigenvalues and connect different scales with explicit coordinate constructions, such as degree-12 polynomial spectral filters and dihedral character projectors, and they contain lemmas that the final inequality does not use. Automated provers work by local symbolic rewriting in a fixed basis. Without the geometric pictures that physicists use for many-body systems, such a search tends to build up intermediate algebra in order to stay correct.

In this work, the authors asked GPT-6 to re-examine the argument in terms of spacetime tensor networks, and worked through the resulting structure interactively. The diagrams make visible the geometric structure that coordinate expansions hide. With imaginary time and space treated alike on a torus, the self-adjoint spatial transfer operator and the doubling iteration reduce to short identities between planar diagrams. This suggests that tensor-network diagrams are useful not only for calculation but also for reading, simplifying, and extracting the physical mechanism of a machine-generated proof. The authors checked all mathematical arguments and formal proofs.

[1] OpenAI, *The periodic spin-one Haldane gap*, manuscript dated September 24, 2026, Proposition 5.5 and Appendices A–C.

[2] OpenAI, *A boundary-field gap for the spin-one Heisenberg chain*, manuscript dated September 24, 2026, Secs. 2–6 and Appendices A and B.

[3] W. Tang, H.-H. Tu, and L. Wang, Continuous matrix product operator approach to finite temperature quantum states, Phys. Rev. Lett. **125**, 170604 (2020).

[4] B. Pirvu, V. Murg, J. I. Cirac, and F. Verstraete, Matrix product operator representations, New J. Phys. **12**, 025012 (2010).

[5] Companion note, *Scope of the spacetime bootstrap* (`thermal-mpo-gap-scope.tex`, 2026), Appendix E.

[6] T. Kennedy, Exact diagonalisations of open spin-1 chains, J. Phys.: Condens. Matter **2**, 5737 (1990).

Appendix A: Open chain with boundary fields

The networks above also bound the gap of the open chain with equal endpoint fields. On n spin-1 sites, with $\sum_\alpha (S^\alpha)^2 = 2I$, let

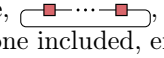
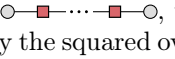
$$H_n^{\mathrm{s}} = \sum_{i=1}^{n-1} \boldsymbol{S}_i \cdot \boldsymbol{S}_{i+1} - h(S_1^z + S_n^z), \qquad h = \frac{3}{5}. \tag{A1}$$

Theorem 2 (Boundary-field gap [2])**.** *For every odd $n \geq 1921$, the Hamiltonian (A1) has a unique ground state and $\Delta_n > (\log 10)/392 \simeq 0.005874$.*

Boundary fields are essential here: without them, each end of the open spin-1 chain carries an effective spin $\frac{1}{2}$, and the four lowest states are split only by an amount exponentially small in n [6]. As on the ring, the gap follows from the thermal purity: by Eqs. (2) and (3), it suffices to bound the equal-field thermal purity $P_\tau^{\mathrm{s}}(n,\beta) = \sum_m w_m^2$ from below, since

$$e^{-\beta\Delta_n} = \frac{w_1}{w_0} \leq \frac{1 - P_\tau^{\mathrm{s}}(n,\beta)}{P_\tau^{\mathrm{s}}(n,\beta)}. \tag{A2}$$

Strategy.— The ring proof cannot be applied directly, because an open chain has no ring to close: its partition function is not a trace of X_β^n, so there is no spatial purity to square. Its bulk, however, is the same as that of the ring, and the ring estimates already control the spectrum of X_β. We therefore keep the bulk tensors and put everything new into the two ends. On a ring, every virtual leg of Eq. (7) joins two tensors; on an open chain, the outer legs of the first and last sites have no partner, and a vector on the virtual space, a *boundary vector*, must be inserted there. This is the quantum counterpart of the open classical Ising chain, whose partition function is $u^{\mathsf{T}} K^{n-1} u$, with $u = (e^{\beta h}, e^{-\beta h})$ for a field h on each end spin.

The boundary vectors change which eigenvectors of X_β the network sees. A periodic network is closed by a trace,  , in which every eigenvector, the leading one included, enters with weight one. An open chain is closed by two boundary vectors,  , which weight the leading eigenvalue $\lambda(\beta)$ by the squared overlap $c(\beta)$ of the leading eigenvector with the boundary vector [Eq. (A9)]. For long chains the open partition function is therefore approximately $c(\beta)\lambda(\beta)^n$, and its thermal purity is approximately $[\lambda(2\beta)/\lambda(\beta)^2]^n\, c(2\beta)/c(\beta)^2$: a bulk factor, controlled by the ring, times a boundary factor independent of n. The boundary factor cancels when the purities at lengths M and $2M$ are compared, so doubling the length costs only the bulk factor; squaring in imaginary time, Eq. (11), needs no ring at all. These two moves replace the spatial squaring of the main text. The approximation requires that the leading term dominates the remainder, which fails if the boundary vector is nearly orthogonal to the leading eigenvector. The proof therefore tracks this *boundary overlap* alongside the thermal purity, through a boundary defect e that shrinks as the chain grows. One step of the resulting induction is three moves: control the overlap, double the length, and square in imaginary time (Scale induction, below).

Scales.— The scales of this appendix replace those of Eq. (19):

$$\beta_j = 49\,2^{j-2}, \quad n_j = 120\,2^{j-2}, \quad M_j = 2n_j \quad (j \geq 2), \tag{A3}$$

with the lengths $k_2 = 142$ and $k_{j+1} = M_j$ at which the boundary defect is evaluated. The ring estimates at these scales, with ring purity defect ε_j, are collected in Appendix B [Eq. (B1)], and the starting values at $j = 2$ in Appendix C.

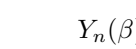
Boundary vectors.— A field on an end spin enters the network as half a bond, whose events insert G_z on the end site and a number outside it. For a history a of d bond events, the boundary vector is

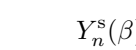
$$v_\beta(a) = \begin{cases} (ih)^d, & \text{all labels are } z, \\ 0, & \text{otherwise,} \end{cases} \qquad \|v_\beta\|^2 = e^{\beta h^2}, \tag{A4}$$

with value one on the empty history; the norm sums $(h^2\beta)^d/d!$ over d. Each event on a cap inserts a field on the end site: the left cap is conjugated, so its factor $(-ih)G_z = +hS^z$ adds $-hS_1^z$ to H, while the same vector on the right, $(ih)G_z = -hS^z$, adds $+hS_n^z$. Identical caps thus give opposite end fields, the chain $H_n^{\mathrm{o}} = H_n^{\mathrm{s}} + 2hS_n^z$,  . The π rotation $P = \mathrm{diag}(1,-1,-1)$ about x flips every z label, so its virtual action U_P reverses the right field and restores equal fields,  . The open chain uses the tensor W_β of Eq. (5), with the energy shift $\kappa = 700741/500000$ of Ref. [2] (called a there) absorbed as a factor $e^{-\kappa\beta}$, i.e. $H_n^\nu \to H_n^\nu + \kappa n I$; it cancels from every Gibbs state and every purity. Opening the ring of Eq. (7) and capping it with boundary vectors gives

$$e^{-\beta H_n^{\mathrm{s}}} = \text{  }. \tag{A5}$$

The right cap alone therefore sets the field pattern of the trace:

$$Y_n(\beta) = \mathrm{Tr}\, e^{-\beta H_n^{\mathrm{o}}} = \text{  }, \tag{A6}$$

$$Y_n^{\mathrm{s}}(\beta) = \mathrm{Tr}\, e^{-\beta H_n^{\mathrm{s}}} = \text{  }, \tag{A7}$$

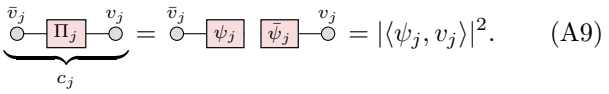
where X_β^n is the chain of Eq. (9), and Z_n of Eq. (8) is the periodic counterpart. The opposite-field chain has identical bra and ket caps, so its even moments are nonnegative; this is why the proof works with H_n^{o} and transfers to H_n^{s} only at the end. Simultaneous spin rotations of the histories act on $L^2(\mu_\beta)$ as real orthogonal operators commuting with X_β; in particular $[U_P, X_\beta] = 0$. Before the shift, the absolute traced terms sum to at most $3^n e^{\beta[3(n-1)+2h]}$, so the exponential series is absolutely convergent and may be regrouped into the contraction of Eq. (A5).

Transfer operator spectrum.— The open-chain argument needs the following facts about X_β, all supplied by the ring. At successive inverse temperatures $\beta_{j+1} = 2\beta_j$, the ring estimates of Appendix B show that the leading eigenvalue of Eq. (8), $\lambda_j := \lambda_0(\beta_j)$, is simple and positive. Normalizing it to one:

$$T_j := X_{\beta_j}/\lambda_j = \Pi_j + R_j, \qquad \Pi_j = |\psi_j\rangle\langle\psi_j|, \quad \|\psi_j\| = 1, \tag{A8}$$

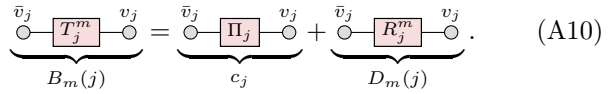
where $\Pi_j R_j = R_j \Pi_j = 0$, so $T_j^m = \Pi_j + R_j^m$. Furthermore, $\|R_j\| < 1$, $U_P \Pi_j = \Pi_j$, and $0 < r_j := \lambda_{j+1}/\lambda_j^2 \le 1$ (Appendix B).

Boundary overlap and residual decomposition.— We now split the open-chain contraction into the leading part, which carries the boundary overlap, and a remainder. Put $v_j = v_{\beta_j}$. Inserting the leading projector disconnects the network:



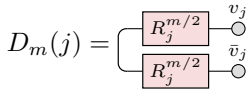
$$= |\langle\psi_j, v_j\rangle|^2. \tag{A9}$$

The overlap c_j is the squared projection of v_j onto ψ_j, ; unlike a periodic trace, where $\mathrm{Tr_v}\,\Pi_j = 1$, it vanishes when the two are orthogonal. For even $m \ge 2$,



. (A10)

The remainder is nonnegative: writing $R_j^m = R_j^{m/2} R_j^{m/2}$ folds the chain at its middle into a norm,



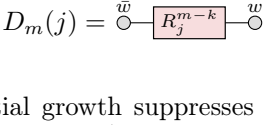
$$= \|R_j^{m/2} v_j\|^2 \ge 0. \tag{A11}$$

Self-adjointness of R_j makes the lower row the mirror of the upper one, while $B_m = Y_m/\lambda_j^m > 0$. The boundary defect is defined by

$$e_m(j) := \frac{D_m(j)}{B_m(j)} = 1 - \frac{c_j}{B_m(j)}, \tag{A12}$$

the grey fraction of the bar, , drawn at the starting value $e_{142}(2) \le .14$ of Eq. (C1). Any $e_m < 1$ guarantees a nonzero overlap with the leading eigenvector.

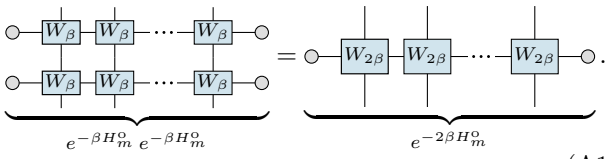
Spatial remainder suppression.— A longer chain shrinks the remainder but leaves the overlap unchanged. For even $m \ge k$, moving $k/2$ factors of R_j onto each boundary vector gives $w = R_j^{k/2} v_j$ and



$$D_m(j) = \le \|R_j\|^{m-k} \underbrace{\|w\|^2}_{D_k(j)}. \tag{A13}$$

Spatial growth suppresses the remainder geometrically, by $\|R_j\|^{m-k}$, while leaving c_j invariant, ensuring $c_j \le B_m(j) \le B_k(j)$.

Thermal operator composition.— Composition in imaginary time, $e^{-\beta H_m^o} e^{-\beta H_m^o} = e^{-2\beta H_m^o}$, couples temperatures:



. (A14)

This is an identity between operators on the physical Hilbert space; it does not assert that two W_β tensors fuse into one $W_{2\beta}$ site by site. Normalizing the trace defines the opposite-field thermal purity:

$$P_\tau^o(m, \beta_j) := \frac{Y_m(\beta_{j+1})}{Y_m(\beta_j)^2} = \mathrm{Tr}\,\rho_m^o(\beta_j)^2 \in (0, 1], \tag{A15}$$

with $\rho_m^o(\beta) = e^{-\beta H_m^o}/Y_m(\beta)$. Since $Y_m(\beta_{j+1}) = P_\tau^o(m, \beta_j) Y_m(\beta_j)^2$ and $B_m(j) = Y_m(\beta_j)/\lambda_j^m$, factoring out leading eigenvalues links adjacent temperatures:

$$B_m(j+1) = P_\tau^o(m, \beta_j) r_j^{-m} B_m(j)^2. \tag{A16}$$

Coupled overlap recurrence.— This is Move 1: a small boundary defect at scale j gives a small one at scale $j+1$. Assuming $e_k(j) < 1$ at scale j, let primes denote $j+1$, choose even $m > k \ge 2$, and let $\|R_{j+1}\|^{m-k} \le q < 1$. Spatial suppression at $j+1$ gives $D'_m \le q D'_k$. Since c' is length-independent, $D'_k - D'_m = B'_k - B'_m$; substituting $D'_k = D'_m + B'_k - B'_m$ into $D'_m \le q D'_k$ gives the first inequality, and dividing by B'_m the second:

$$(1-q) D'_m \le q(B'_k - B'_m), \qquad e'_m \le \frac{q}{1-q}\left(\frac{B'_k}{B'_m} - 1\right). \tag{A17}$$

Composition relates the ratio to the previous temperature:

$$\frac{B'_k}{B'_m} = \frac{P_\tau^o(k, \beta_j)}{P_\tau^o(m, \beta_j)} \left[\frac{B_k(j)}{B_m(j)}\right]^2 r_j^{m-k} \le \frac{1}{P_\tau^o(m, \beta_j)[1 - e_k(j)]^2}, \tag{A18}$$

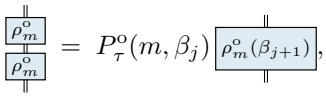
using $P_\tau^o(k, \beta_j), r_j \le 1$ and $B_m(j) \ge c_j = [1 - e_k(j)] B_k(j) > 0$. Because only the previous overlap appears in the denominator, $e'_m < 1$ establishes that the updated overlap remains positive. When $e_k(j)$ and $1 - P_\tau^o(m, \beta_j)$ are $\mathcal{O}(x)$, choosing $m - k$ with $q \le \mathcal{O}(x)$ ensures that e'_m contracts quadratically as $\mathcal{O}(x^2)$.

Thermal purity propagation.— This is Move 2: with the overlap under control, the length can double at almost no cost in purity. For even M, dividing Eq. (A16) at $m = 2M$ by the same equation at $m = M$ yields

$$\frac{P_\tau^o(2M, \beta_j)}{P_\tau^o(M, \beta_j)} = r_j^M \underbrace{\frac{B_{2M}(j+1)}{B_M(j+1)}}_{\ge 1 - e_M(j+1)} \underbrace{\left[\frac{B_M(j)}{B_{2M}(j)}\right]^2}_{\ge 1}. \tag{A19}$$

The first bound retains c_{j+1}, while the second uses Eq. (A13). Because $r_j^M \approx 1$ [Eq. (B2)], purity remains controlled upon doubling length. Squaring the Gibbs op-

erator advances temperature, as in Eq. (11):



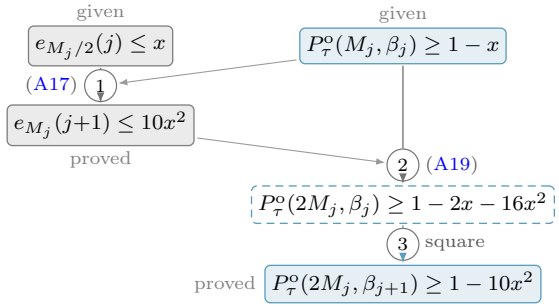
$$= P_\tau^{\mathrm{o}}(m,\beta_j)\,\rho_m^{\mathrm{o}}(\beta_{j+1}), \tag{A20}$$

so the squaring bound (12) at $m = 2M$ drives $1 - P_\tau^{\mathrm{o}}$ to $\mathcal{O}(x^2)$.

Scale induction.— Together, the overlap and purity bounds give an induction in which the length and the inverse temperature double at each step. With the scales (A3), the lengths k_j, and the ring purity defect ε_j bounded in Eq. (B1), induction establishes

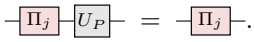
$$\begin{aligned} &\varepsilon_j,\ e_{k_j}(j),\ 1 - P_\tau^{\mathrm{o}}(M_j,\beta_j) \le x_j \quad (j \ge 5),\\ &x_5 = .01, \qquad x_{j+1} = 10x_j^2, \qquad x_j = 10^{-1-2^{j-5}}. \end{aligned} \tag{A21}$$

For $j \ge 3$, $k_j = M_j/2$, leaving $M_j/2$ sites to suppress remainders. One step $j \to j+1$ is three moves. In the diagram below, read from top to bottom, each box is one bound: grey for the boundary defect $e_k(j)$, blue for the opposite-field thermal purity P_τ^{o}. Solid boxes are given at scale j or proved at scale $j+1$, the dashed box is intermediate, and the grey arrow entering a move is the second bound it uses; the bounds shown hold for $j \ge 5$ (Appendix C):



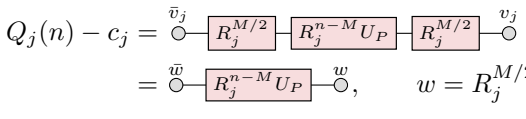
Move 1 is the overlap recurrence (A17)–(A18), move 2 the length doubling (A19), and move 3 the squaring (A20). Since $M_{j+1} = 2M_j$, the proved pair is the given pair of scale $j+1$, so the step repeats. The defect falls doubly exponentially, $x_5 = 10^{-2} \to x_9 = 10^{-17}$ (log scale); since $\beta_j = 392\,2^{j-5}$, it decays exponentially in inverse temperature.

Equal boundary fields and odd lengths.— The opposite-field estimates extend to the equal-field Hamiltonian (A1) via the spin-rotation symmetry U_P. Next to the leading projector, the rotation operator acts trivially:



$$= \quad . \tag{A22}$$

Thus $Q_j(n) := Y_n^{\mathrm{s}}(\beta_j)/\lambda_j^n$ shares the leading value c_j. For any even $M \le n$, commuting U_P through the right-hand tensors ($[U_P, R_j] = 0$) isolates the remainder:



$$w = R_j^{M/2} v_j. \tag{A23}$$

Because $\|R_j^{n-M} U_P\| \le 1$,

$$|Q_j(n) - c_j| \le \|w\|^2 = D_M(j), \tag{A24}$$

bounding the odd-length remainder by $D_M(j)$ without requiring positivity of R_j^n.

For $M_j \le n \le 2M_j$ and $M = M_j = k_{j+1}$, Eq. (A24) gives $Q \le c + D_M = B_M$ and $Q \ge c - D_M = (1 - 2e_M)B_M$; with $e_M(j+1) \le x_{j+1}$ from Eq. (A21), this yields

$$Q_j(n) \le B_M(j), \quad Q_{j+1}(n) \ge (1 - 2x_{j+1})B_M(j+1) > 0. \tag{A25}$$

Since $P_\tau^{\mathrm{s}}(n,\beta_j) = Q_{j+1}(n)\lambda_{j+1}^n / [Q_j(n)\lambda_j^n]^2$, inserting these two bounds and Eq. (A16) at $m = M$ gives the equal-field purity:

$$\begin{aligned} P_\tau^{\mathrm{s}}(n,\beta_j) = \frac{Y_n^{\mathrm{s}}(\beta_{j+1})}{Y_n^{\mathrm{s}}(\beta_j)^2} &\ge (1 - 2x_{j+1}) P_\tau^{\mathrm{o}}(M,\beta_j) r_j^{n-M} \\ &\ge (1 - 2x_{j+1})(1 - x_j)(1 - \varepsilon_j)(1 - \varepsilon_{j+1}) \\ &\ge 1 - 2x_j - 26x_j^2 \ge 1 - 4x_j, \end{aligned} \tag{A26}$$

using $r_j^{n-M} \ge r_j^M$ [Eq. (B2)] and Appendix C bounds ($\varepsilon_j \le x_j$, $\varepsilon_{j+1} \le 6x_j^2$, $x_{j+1} = 10x_j^2$).

Proof of Theorem 2. The intervals $[M_j, 2M_j]$ cover every $n \ge M_5 = 1920$. For odd $n \ge 1921$, choose such a j. The largest Gibbs weight satisfies $w_0 \ge P_\tau^{\mathrm{s}}(n,\beta_j) \ge 1 - 4x_j > 1/2$: the spectrum looks like ▙, so the ground state is unique. Equation (A2) gives

$$e^{-\beta_j \Delta_n} \le \frac{4x_j}{1 - 4x_j} \le \frac{5}{12}\, 10^{-2^{j-5}} < 10^{-2^{j-5}}. \tag{A27}$$

Taking logarithms and using $\beta_j = 392\,2^{j-5}$ yields $\Delta_n > (\log 10)/392$. □

Appendix B: Transfer bounds at the open-chain scales

For the scales in Eq. (A3), Appendix A uses

$$\begin{aligned} &\|R_j\|^{n_j} \le d(\varepsilon_j), \qquad d(\varepsilon) = \frac{\varepsilon}{2 - 3\varepsilon},\\ &\varepsilon_2 = .052, \qquad \varepsilon_{j+1} = 6\varepsilon_j^2, \end{aligned} \tag{B1}$$

and

$$0 < r_j \le 1, \quad r_j^{M_j} \ge (1 - \varepsilon_j)(1 - \varepsilon_{j+1}), \quad U_P \Pi_j = \Pi_j. \tag{B2}$$

These follow from the first row of Eq. (C1) by the doubling estimate of the main text, with a larger starting defect: if $P_{\mathrm{s}}(n,\beta), P_\tau(2n,\beta) \ge 1 - \varepsilon$, move 1 of Eq. (16) gives $P_{\mathrm{s}}(n,2\beta) \ge (1 - \varepsilon)^3$, and squaring with Eq. (12) bounds $1 - P_{\mathrm{s}}(2n,2\beta)$ by

$$f(1 - (1 - \varepsilon)^3) = \frac{\varepsilon^2}{2}\left[\sum_{s=1}^{3} (1 - \varepsilon)^{-s}\right]^2 \le 6\varepsilon^2 \tag{B3}$$

for $0 \le \varepsilon \le .052$, in place of $5\varepsilon^2$. Since $6\varepsilon^2 \le \varepsilon$, moves 3 and 4 give the same bound for $1 - P_\tau(4n, 2\beta)$. Induction therefore proves $P_{\rm s}(n_j, \beta_j), P_\tau(M_j, \beta_j) \ge 1 - \varepsilon_j$.

To turn a purity bound into a bound on the remainder R_j, we use an elementary fact about probability lists. Consider a probability list with purity at least $1-\varepsilon > 1/2$ and largest weight p_0; put $z = 1 - p_0$. The purity is at most $p_0^2 + z^2 = 1 - 2z(1-z)$, so $2z(1-z) \le \varepsilon$. It is also at most p_0, so $p_0 > 1/2$: the largest weight is unique and $z < 1/2$. Then $z \le 2z(1-z) \le \varepsilon$, hence $z \le \varepsilon/[2(1-\varepsilon)]$ and $z/(1-z) \le d(\varepsilon)$. Applied to the eigenvalues $p_i = |\lambda_i(\beta_j)|^{n_j}/Z_{n_j}(\beta_j)$ of $\sigma_{n_j}(\beta_j)$, the largest-modulus eigenvalue is simple and

$$\sum_{i \ge 1} \left| \frac{\lambda_i(\beta_j)}{\lambda_j} \right|^{n_j} \le d(\varepsilon_j), \tag{B4}$$

which implies Eq. (B1). Positivity of $Z_N(\beta_j)$ for all large odd N ensures that the leading eigenvalue is positive, since a negative leading eigenvalue would make $\mathrm{Tr_v}\, X_{\beta_j}^N$ negative. The residual sum tends to zero since X_{β_j} is Hilbert–Schmidt and its residual spectral radius is strictly smaller than λ_j.

It remains to show $U_P \Pi_j = \Pi_j$. The real kernel admits real orthogonal virtual actions of the three π rotations and the cyclic permutation of the coordinate axes, all commuting with X_{β_j}. On its real one-dimensional leading eigenspace these actions are signs. The order-three cyclic rotation has sign $+1$ and conjugates the three π rotations into one another, so their signs agree. The product of two is the third; their common sign is therefore $+1$. This proves $U_P \Pi_j = \Pi_j$.

Finally $Z_N(\beta_{j+1}) \le Z_N(\beta_j)^2$ since the thermal purity $P_\tau(N, \beta_j)$ is at most one. Taking Nth roots along even $N \to \infty$ gives $\lambda_{j+1} \le \lambda_j^2$. For the lower bound at the required finite length, the largest eigenvalue of $\sigma_{M_j}(\beta_{j+1})$ is at least $1 - \varepsilon_{j+1}$, since $M_j = n_{j+1}$. Hence

$$\begin{aligned} \lambda_{j+1}^{M_j} &\ge (1 - \varepsilon_{j+1}) Z_{M_j}(\beta_{j+1}) \\ &\ge (1 - \varepsilon_{j+1})(1 - \varepsilon_j) Z_{M_j}(\beta_j)^2 \\ &\ge (1 - \varepsilon_{j+1})(1 - \varepsilon_j) \lambda_j^{2M_j}, \end{aligned}$$

which is the remaining bound in Eq. (B2).

Appendix C: Starting estimates for the open chain

The induction of Appendix A [Eq. (A21)] starts at $j = 5$ and needs all three defects below $x_5 = .01$ there. The certified inputs live at $j = 2$ and are much larger. This appendix closes the gap in four steps: it states the inputs, turns the three moves of Appendix A into update rules for numerical bounds, runs these rules from $j = 2$ to $j = 5$, and finally proves the induction step for $j \ge 5$.

Inputs.— The certified starting estimates combine periodic variational estimates with small-cluster boundary certificates from Ref. [2]:

$$\begin{aligned} P_{\rm s}(120, 49) \ge .948, \qquad & P_\tau(240, 49) \ge .948, \\ e_{142}(2) \le .14, \qquad & P_\tau^{\rm o}(240, \beta_2) \ge .79. \end{aligned} \tag{C1}$$

The first row concerns the ring at the scale $(n_2, \beta_2) = (120, 49)$ of Eq. (A3); it gives the ring purity defect $\varepsilon_2 = .052$ used in Appendix B. The second row concerns the open chain: the boundary defect at length $k_2 = 142$ and the opposite-field thermal purity at length $M_2 = 240$.

Update rules.— Write E_j for an upper bound on the boundary defect $e_{k_j}(j)$ and η_j for an upper bound on the thermal defect $1 - P_\tau^{\rm o}(M_j, \beta_j)$. Each move of Appendix A becomes a rule that computes the bounds at scale $j+1$ from those at scale j. Move 1 controls the overlap. Combining Eqs. (A17) and (A18) gives

$$\begin{aligned} e_m(j+1) &\le F(q, e_k(j), P_\tau^{\rm o}(m, \beta_j)), \\ F(q, e, t) &= \frac{q}{1-q} \left[\frac{1}{(1-e)^2 t} - 1 \right]. \end{aligned} \tag{C2}$$

This function increases with q, e and decreases with t, so bounds may be inserted for its arguments. The suppression factor q must bound $\|R_{j+1}\|^{M_j - k_j}$; Eq. (B1) bounds $\|R_{j+1}\|^{n_{j+1}}$, and $n_{j+1} = M_j$, which fixes the exponent in Eq. (C3) below. Moves 2 and 3 control the purity: the length doubling (A19) multiplies four factors, namely the purity $1 - \eta_j$ at length M_j, the bulk ratio $r_j^{M_j} \ge (1 - \varepsilon_j)(1 - \varepsilon_{j+1})$ of Eq. (B2), and the overlap factor $1 - E_{j+1}$, and the squaring bound (12) then applies f. With $k_2 = 142$ and $k_{j+1} = M_j$, the updates are

$$q_j = d(\varepsilon_{j+1})^{(M_j - k_j)/M_j}, \tag{C3}$$

$$E_{j+1} \ge F(q_j, E_j, 1 - \eta_j), \tag{C4}$$

$$\eta_{j+1} \ge f\big(1 - (1 - \eta_j)(1 - \varepsilon_j)(1 - \varepsilon_{j+1})(1 - E_{j+1})\big). \tag{C5}$$

Any E_{j+1}, η_{j+1} in $[0, 1)$ satisfying these inequalities are valid bounds at scale $j+1$.

Three explicit steps.— The inputs (C1) give $\varepsilon_2 = .052$, $E_2 = .14$, and $\eta_2 = 1 - .79 = .21$. Three applications of Eqs. (C4)–(C5) give the following rational upper bounds:

j	E_j	η_j	q_j (upper bound)
2	.14	.21	.142
3	.12	.155	.0282
4	.016	.026	.0028
5	.001	.001	—

Both defects collapse along the iteration, E_j:  and η_j:  ($j = 2, \ldots, 5$, each relative to $j = 2$). Rational arithmetic confirms these step bounds, with subsequent steps verified by squaring, and $\varepsilon_5 < .000015$. For example, at $j = 2$ the exponent in Eq. (C3) is $(240 - 142)/240 = 49/120$, so $q_2 \le .142$ amounts to $d(\varepsilon_3)^{49} < .142^{120}$. At $j = 5$ all three defects are below .01, which is the base case of Eq. (A21).

The induction step.— Induction on scale verifies Eq. (A21) for all subsequent scales. Assuming the bound at scale j with $x = x_j \leq .01$, the relation $k_j = M_j/2$ gives

$$q_j \leq \sqrt{\frac{6x^2}{2-18x^2}} \leq 2x, \qquad F(2x, x, 1-x) \leq \frac{6x^2}{(1-2x)(1-x)^3} < 7x^2. \tag{C6}$$

Setting $E_{j+1} = 10x^2$ satisfies Eq. (C4). The four-factor product in Eq. (C5) is at least $1-2x-16x^2$, and $f(2x+16x^2) < 3x^2$, so $\eta_{j+1} = 10x^2$ satisfies Eq. (C5) with $\varepsilon_{j+1} \leq 6x^2$, completing the induction for Eq. (A21).